

Question

For aqueous solutions containing iodide ions (excluding other ions that may affect the determination), the following method can be used to determine the content of iodide ions:

1. Under neutral or weakly acidic conditions, add excess Br_2 for oxidation. After removing the remaining Br_2 , add excess KI ;
2. Extract with CCl_4 (approximately considered complete extraction). After separating the aqueous phase, back-extract with an aqueous solution of N_2H_4 ;
3. Under neutral or weakly acidic conditions, oxidize with excess Br_2 . After removing the remaining Br_2 , add excess KI ;
4. Appropriately acidify the solution, use **X** as an indicator, and titrate with **Y** standard solution.

In a certain determination, 25.00 mL of the test solution containing KI was taken. After the above operations, 20.06 mL of 0.1000 mol/L **Y** standard solution was consumed.

Which of the following options is correct?

- A.** All other options are incorrect
- B.** **X** should be sodium diphenylamine sulfonate, and **Y** should be sodium thiosulfate.
- C.** Under neutral or weakly acidic conditions, the purpose of adding excess Br_2 is to oxidize I^- to I_2
- D.** The target substance for back extraction using an aqueous solution of N_2H_4 is I^- in the organic phase
- E.** The concentration of I^- in the original solution is 13.37 mmol/L.
- F.** The concentration of I^- in the original solution is 2.229 mmol/L.

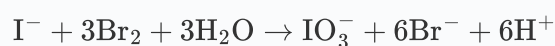
G. The concentration of I^- in the original solution is 80.24 mmol/L.

Answer

Correct Answer: **F**

Detailed Explanation

Under neutral or weakly acidic conditions, the addition of excess Br_2 can oxidize I^- to IO_3^- :



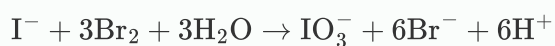
CHECKPOINT

1 PTS

Br_2 can oxidize I^- to IO_3^-

CHECKPOINT

1 PTS

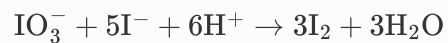


After removing the remaining Br_2 , adding excess KI can reduce IO_3^- to I_2 :



CHECKPOINT

1 PTS



After extracting I_2 into the organic phase with CCl_4 , back extraction is performed with an aqueous solution of N_2H_4 to return I_2 to the aqueous phase in the form of I^- :

CHECKPOINT

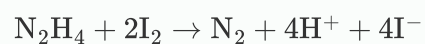
1 PTS

Back extraction is performed with an aqueous solution of N_2H_4 to return I_2 to the aqueous phase in the form of I^-



CHECKPOINT

1 PTS



The oxidation extraction process in steps 1 and 2 is equivalent to amplifying the content of I^- by 6 times:



CHECKPOINT

1 PTS

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The repeated operations in steps 3 and 4 once again amplify the concentration of I^- by 6 times:



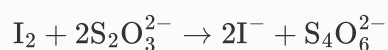
Therefore, this method totally amplifies the initial iodide ion concentration by 36 times.

CHECKPOINT

1 PTS

This method totally amplifies the initial iodide ion concentration by 36 times

The content of I_2 should be titrated using starch as an indicator (**X**) and $Na_2S_2O_3$ as a titrant (**Y**), with the titration relationship being:



CHECKPOINT

1 PTS

X is starch and **Y** is $Na_2S_2O_3$

Therefore, the titration consumption of $\text{Na}_2\text{S}_2\text{O}_3$ is:

$$0.1000 \times 20.06 = 2.006 \text{ mmol}$$

Corresponding to I^- in the original solution:

$$\frac{1}{36} \times 2.006 = 5.572 \times 10^{-2} \text{ mmol}$$

Then the concentration of I^- in the original solution is:

$$5.572 \times 10^{-2} / \frac{25.00}{1000} = 2.229 \text{ mmol/L}$$

CHECKPOINT

2 PTS

The concentration of I^- in the original solution is 2.229 mmol/L